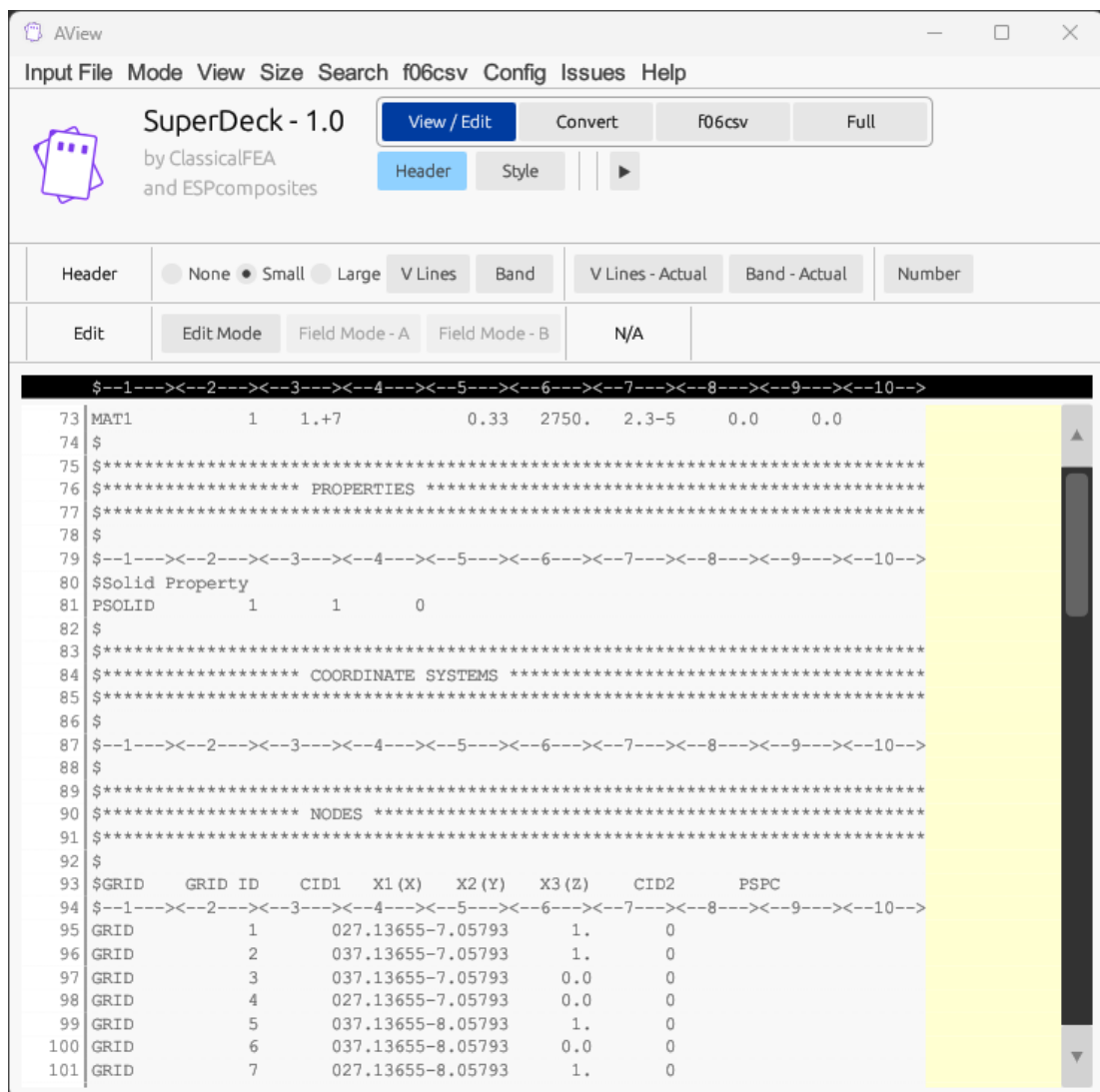


SuperDeck

- View
- Edit
- Convert Format
- f06csv – Parse f06 file to a CSV

Basic approach where the small format is manually chosen...



VIEW

This time, the **V Lines** and **Band** options are selected.

AView

Input File

Mode

View

Size

Search

f06csv

Config

Issues

Help

SuperDeck - 1.0

by ClassicalFEA
and ESPcomposites

View / Edit

Convert

f06csv

Full

Header

Style

Header

None

Small

Large

V Lines

Band

V Lines - Actual

Band - Actual

Number

Edit

Edit Mode

Field Mode - A

Field Mode - B

N/A

\$--1---<--2---<--3---<--4---<--5---<--6---<--7---<--8---<--9---<--10-->

73

MAT1

1

1.+7

0.33

2750.

2.3-5

0.0

0.0

74

\$

75

\$

76

\$

***** PROPERTIES *****

77

\$

78

\$

79

\$

80

\$Solid Property

81

PSOLID

1

1

0

82

\$

83

\$

84

\$

***** COORDINATE SYSTEMS *****

85

\$

86

\$

87

\$

88

\$

89

\$

90

\$

***** NODES *****

91

\$

92

\$

93

\$GRID

GRID ID

CID1

X1 (X)

X2 (Y)

X3 (Z)

CID2

PSPC

94

\$

95

GRID

1

027.13655-7.05793

1.

0

96

GRID

2

037.13655-7.05793

1.

0

97

GRID

3

037.13655-7.05793

0.0

0

98

GRID

4

027.13655-7.05793

0.0

0

99

GRID

5

037.13655-8.05793

1.

0

100

GRID

6

037.13655-8.05793

0.0

0

101

GRID

7

027.13655-8.05793

1.

0

VIEW

In this case, **V Lines – Actual** and **Band – Actual** are used. This may be useful for decks that have a combination of free, small, and large formats.

Unlike the basic approach, SuperDeck has built in logic to detect what format an entry (card) is.

The screenshot shows the AView SuperDeck - 1.0 application window. The title bar includes 'AView' and standard window controls. The menu bar contains 'Input File Mode View Size Search f06csv Config Issues Help'. The main interface has a toolbar with 'View / Edit', 'Convert', 'f06csv', and 'Full' buttons. Below the toolbar, there are tabs for 'Header', 'Style', and a play button. The main display area is divided into several sections: a 'Header' section with radio buttons for 'None', 'Small', and 'Large', and buttons for 'V Lines', 'Band', 'V Lines - Actual', 'Band - Actual', and 'Number'; an 'Edit' section with buttons for 'Edit Mode', 'Field Mode - A', 'Field Mode - B', and 'N/A'; and a large table of card data. The table has columns for line numbers, card types (e.g., GRID*), and various numerical values. The table is currently displaying a list of cards, with some cells highlighted in yellow. The table structure is as follows:

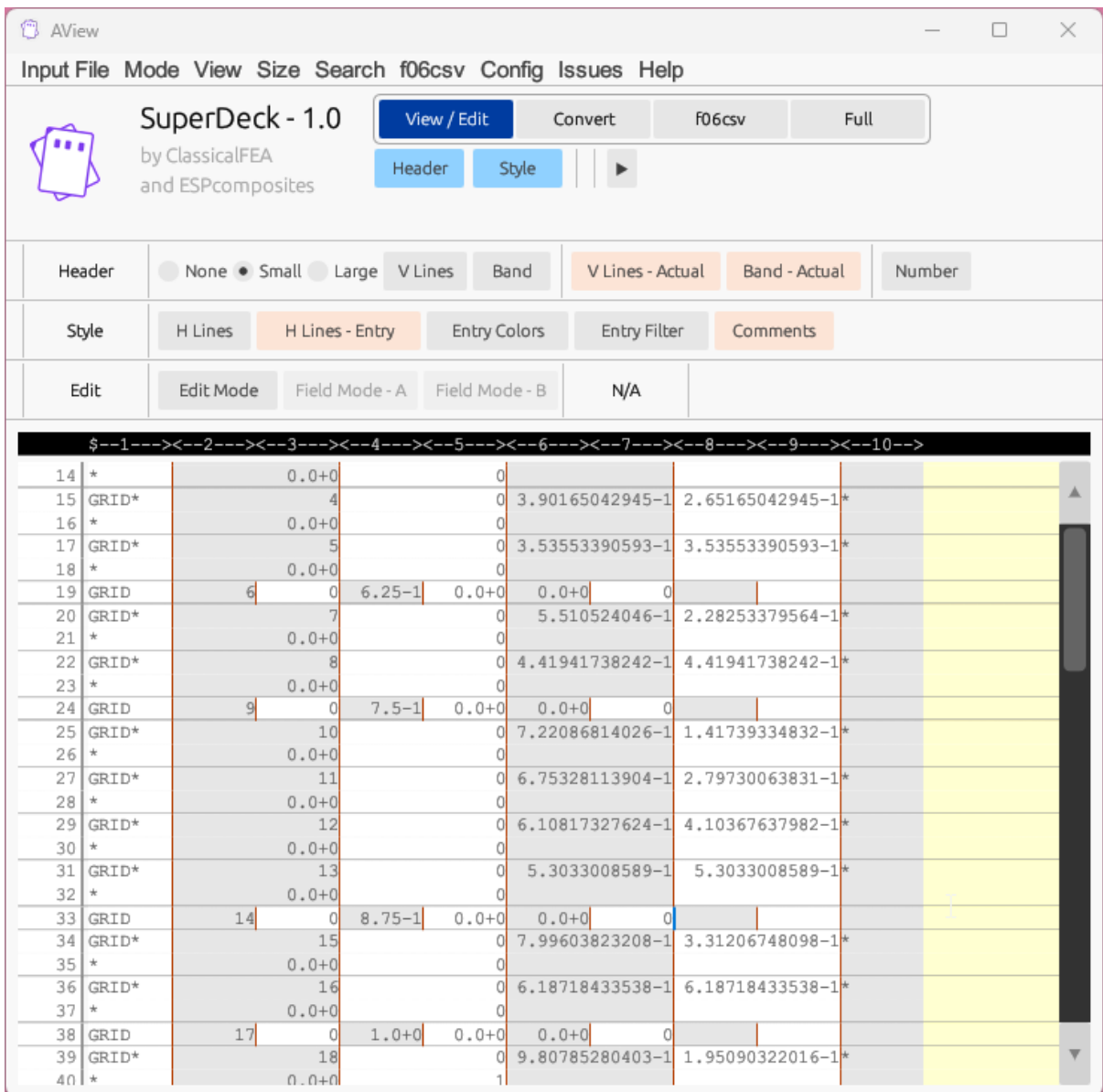
Header	None	Small	Large	V Lines	Band	V Lines - Actual	Band - Actual	Number
Edit	Edit Mode	Field Mode - A	Field Mode - B	N/A				

The table of card data is as follows:

Line	Card Type	Field 1	Field 2	Field 3	Field 4	Field 5	Field 6	Field 7
14	*		0.0+0		0			
15	GRID*		4		0	3.90165042945-1	2.65165042945-1	*
16	*		0.0+0		0			
17	GRID*		5		0	3.53553390593-1	3.53553390593-1	*
18	*		0.0+0		0			
19	GRID	6	0	6.25-1	0.0+0	0.0+0	0	
20	GRID*		7		0	5.510524046-1	2.28253379564-1	*
21	*		0.0+0		0			
22	GRID*		8		0	4.41941738242-1	4.41941738242-1	*
23	*		0.0+0		0			
24	GRID	9	0	7.5-1	0.0+0	0.0+0	0	
25	GRID*		10		0	7.22086814026-1	1.41739334832-1	*
26	*		0.0+0		0			
27	GRID*		11		0	6.75328113904-1	2.79730063831-1	*
28	*		0.0+0		0			
29	GRID*		12		0	6.10817327624-1	4.10367637982-1	*
30	*		0.0+0		0			
31	GRID*		13		0	5.3033008589-1	5.3033008589-1	*
32	*		0.0+0		0			
33	GRID	14	0	8.75-1	0.0+0	0.0+0	0	
34	GRID*		15		0	7.99603823208-1	3.31206748098-1	*
35	*		0.0+0		0			
36	GRID*		16		0	6.18718433538-1	6.18718433538-1	*
37	*		0.0+0		0			
38	GRID	17	0	1.0+0	0.0+0	0.0+0	0	
39	GRID*		18		0	9.80785280403-1	1.95090322016-1	*
40	*		0.0+0		1			
41	GRID*		19		0	9.23879532511-1	3.82683432365-1	*
42	*		0.0+0		2			

VIEW

Further extending on this concept, SuperDeck can detect continuation lines. In this case **H Lines – Entry** is selected, which shows only the horizontal lines between entries.



VIEW

As defined in the config file, each entry can have a color when turned on by the “Entry Colors”. In this case, comments are chosen to be grey to help separate them from active lines.

AVIEW

Input File Mode View Size Search f06csv Config Issues Help

SuperDeck - 1.0
by ClassicalFEA
and ESPcomposites

View / Edit Convert f06csv Full

Header None Small Large V Lines Band V Lines - Actual Band - Actual Number

Style H Lines H Lines - Entry Entry Colors Entry Filter Comments

Edit Edit Mode Field Mode - A Field Mode - B N/A

```
$--1-->--2-->--3-->--4-->--5-->--6-->--7-->--8-->--9-->--10-->
54 $*****
55 $
56 BEGIN BULK
57 $
58 $*****
59 $***** PARAMETERS *****
60 $*****
61 $
62 $The following param is used to create the OP2 binary file for results.
63 PARAM, POST, -1
64 $
65 MAT1 1 1.+7 0.33 2750. 2.3-5 0.0 0.0
66 $
67 $*****
68 $***** PROPERTIES *****
69 $*****
70 $
71 $--1-->--2-->--3-->--4-->--5-->--6-->--7-->--8-->--9-->--10-->
72 $Solid Property
73 PSOLID 1 1 0
74 $
75 $*****
76 $***** NODES *****
77 $*****
78 $
79 $GRID GRID ID CID1 X1 (X) X2 (Y) X3 (Z) CID2 PSPC
80 $--1-->--2-->--3-->--4-->--5-->--6-->--7-->--8-->--9-->--10-->
81 GRID 1 027.13655-7.05793 1. 0
82 GRID 2 037.13655-7.05793 1. 0
83 GRID 3 037.13655-7.05793 0.0 0
84 GRID 4 027.13655-7.05793 0.0 0
85 GRID 5 037.13655-8.05793 1. 0
86 GRID 6 037.13655-8.05793 0.0 0
87 GRID 7 027.13655-8.05793 1. 0
88 GRID 8 027.13655-8.05793 0.0 0
89 GRID 9 027.38655-7.05793 1. 0
90 GRID 10 027.63655-7.05793 1. 0
91 GRID 11 027.83655-7.05793 1. 0
```

VIEW

SuperDeck can identify a field and show the card information and also the associated information for the selected field.

The screenshot displays the AView SuperDeck - 1.0 application window. The interface includes a menu bar (Input File, Mode, View, Size, Search, f06csv, Config, Issues, Help) and a toolbar with buttons for View / Edit, Convert, f06csv, Full, Header, and Style. Below the toolbar, there are tabs for Header, Style, and Edit. The main area shows a data table with columns for GRID ID and CIDE. A pop-up window titled 'Card Info: GRID' is overlaid on the table, providing detailed information about the selected GRID card.

GRID ID	CIDE
1	027
2	037
3	037
4	027
5	037
6	037
7	027
8	027
9	027
10	027
11	027
12	028
13	028
14	028
15	028

Card Info: GRID

Field 4 — X1: X coordinate in CP system (Real or blank)

GRID
Defines the location of a geometric grid point.

All Fields:

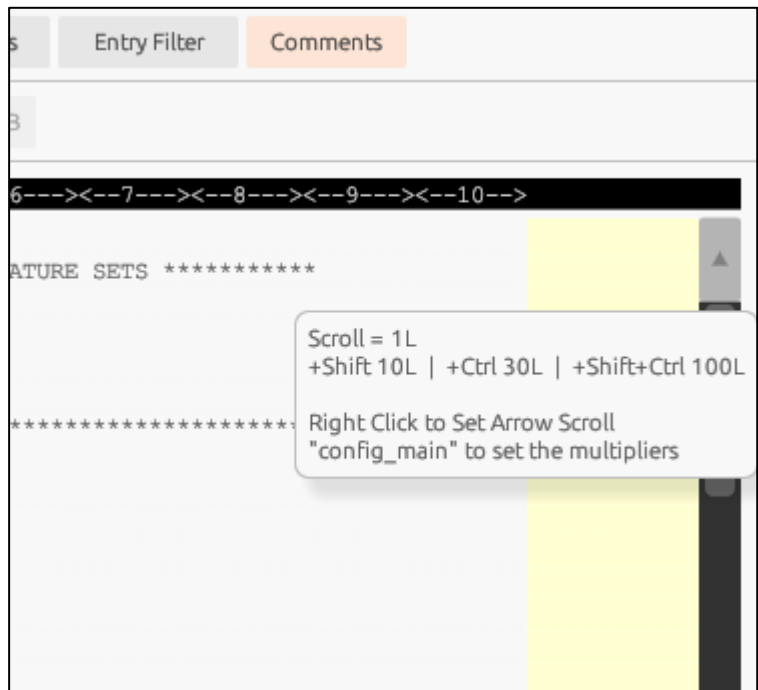
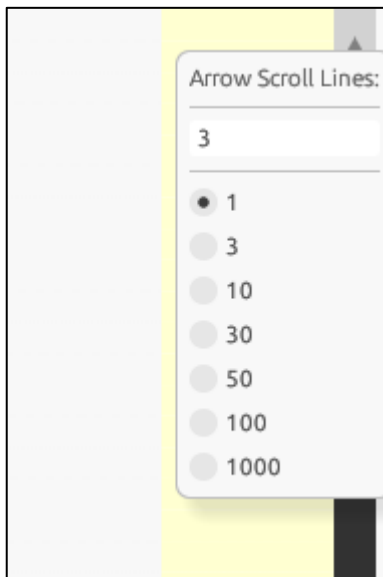
- 2 - ID: Grid point identification number (Integer > 0)
- 3 - CP: Coordinate system ID for location (Integer >= 0 or blank)
- 4 - X1: X coordinate in CP system (Real or blank)
- 5 - X2: Y coordinate in CP system (Real or blank)
- 6 - X3: Z coordinate in CP system (Real or blank)
- 7 - CD: Coordinate system for displacements/forces (Integer >= -1 or blank)

Click outside to close

VIEW

For the scroll arrows at the top/bottom of the scrollbar, the user can define how many rows are moved per click.

There is also an option to modify this magnitude on the fly with keyboard modifiers.



EDIT

SuperDeck has ability to identify and edit fields when an entry is either small or large format.

The screenshot displays the AView SuperDeck - 1.0 application window. The interface includes a menu bar (Input File, Mode, View, Size, Search, f06csv, Config, Issues, Help) and a toolbar with buttons for View / Edit, Convert, f06csv, Full, Header, and Style. Below the toolbar, there are tabs for Header (None, Small, Large), V Lines, Band, V Lines - Actual, Band - Actual, and Number. The Style section has tabs for H Lines, H Lines - Entry, Entry Colors, Entry Filter, and Comments. The Edit section has tabs for Edit Mode, Field Mode - A, Field Mode - B, and N/A.

The main data table is visible, showing columns for GRID, ID, and numerical values. A green box highlights a specific entry in the table, and a dialog box is open for editing that field.

Dialog Box Content:

Small Field (8 chars): 27.13655

Buttons: OK, Cancel

GRID	ID	Value 1	Value 2	Value 3	Value 4	Value 5	Value 6	Value 7	Value 8	Value 9	Value 10
82	GRID	2	037.13655	-7.05793	1.	0					
83	GRID	3	037.13655	-7.05793	0.0	0					
84	GRID	4	027.13655	-7.05793	0.0	0					
85	GRID	5	037.13655	-8.05793	1.	0					
86	GRID	6	037.13655	-8.05793	0.0	0					
87	GRID	7	027.13655	-8.05793	1.	0					
88	GRID	8	0			0					
89	GRID	9									
90	GRID	10	0								
91	GRID	11	027.88655	-7.05793	1.	0					
92	GRID	12	028.13655	-7.05793	1.	0					
93	GRID	13	028.38655	-7.05793	1.	0					
94	GRID	14	028.63655	-7.05793	1.	0					
95	GRID	15	028.88655	-7.05793	1.	0					
96	GRID	16	028.13655	-7.05793	1.	0					
97	GRID	17	029.38655	-7.05793	1.	0					
98	GRID	18	029.63655	-7.05793	1.	0					
99	GRID	19	029.88655	-7.05793	1.	0					
100	GRID	20	030.13655	-7.05793	1.	0					
101	GRID	21	030.38655	-7.05793	1.	0					
102	GRID	22	030.63655	-7.05793	1.	0					
103	GRID	23	010.6	-7.05793	1.	0					
104	GRID	24	031.13655	-7.05793	1.	0					
105	GRID	25	031	-7.05793	1.	0					
106	GRID	26	031.63655	-7.05793	1.	0					

CONVERT

SuperDeck can convert a given deck (dat or bdf) into free, small, or large formats.

Small → Free

SuperDeck - 1.0
by ClassicalFEA and ESPcomposites

Input File (Required): ? Browse... Save Save As Reload C:\AVIEW\target\release\input.dat

Convert

Output File: Browse... C:\AVIEW\target\release\output.dat
Command: aview-cl.exe "C:\AVIEW\target\release\input.dat" "C:\AVIEW\target\release\output.dat" --type=free

Run Free Small Large Stored Comments Warnings Overwrite Orig

GRID ID	GRID ID	CID1	X1 (X)	X2 (Y)	X3 (Z)	CID2	PSPC
79	GRID 1	1	0.0	0.0	0.0	0	0
80	GRID 2	2	0.0	0.0	0.0	0	0
81	GRID 3	3	0.0	0.0	0.0	0	0
82	GRID 4	4	0.0	0.0	0.0	0	0
83	GRID 5	5	0.0	0.0	0.0	0	0
84	GRID 6	6	0.0	0.0	0.0	0	0
85	GRID 7	7	0.0	0.0	0.0	0	0
86	GRID 8	8	0.0	0.0	0.0	0	0
87	GRID 9	9	0.0	0.0	0.0	0	0
88	GRID 10	10	0.0	0.0	0.0	0	0
89	GRID 11	11	0.0	0.0	0.0	0	0
90	GRID 12	12	0.0	0.0	0.0	0	0
91	GRID 13	13	0.0	0.0	0.0	0	0
92	GRID 14	14	0.0	0.0	0.0	0	0
93	GRID 15	15	0.0	0.0	0.0	0	0
94	GRID 16	16	0.0	0.0	0.0	0	0
95	GRID 17	17	0.0	0.0	0.0	0	0
96	GRID 18	18	0.0	0.0	0.0	0	0
97	GRID 19	19	0.0	0.0	0.0	0	0
98	GRID 20	20	0.0	0.0	0.0	0	0
99	GRID 21	21	0.0	0.0	0.0	0	0
100	GRID 22	22	0.0	0.0	0.0	0	0
101	GRID 23	23	0.0	0.0	0.0	0	0
102	GRID 24	24	0.0	0.0	0.0	0	0
103	GRID 25	25	0.0	0.0	0.0	0	0

Small → Large

SuperDeck - 1.0
by ClassicalFEA and ESPcomposites

Input File (Required): ? Browse... Save Save As Reload C:\AVIEW\target\release\input.dat

Convert

Output File: Browse... C:\AVIEW\target\release\output.dat
Command: aview-cl.exe "C:\AVIEW\target\release\input.dat" "C:\AVIEW\target\release\output.dat" --type=large

Run Free Small Large Stored Comments Warnings Overwrite Orig

GRID ID	GRID ID	CID1	X1 (X)	X2 (Y)	X3 (Z)	CID2	PSPC
79	GRID 1	1	0.0	0.0	0.0	0	0
80	GRID 2	2	0.0	0.0	0.0	0	0
81	GRID 3	3	0.0	0.0	0.0	0	0
82	GRID 4	4	0.0	0.0	0.0	0	0
83	GRID 5	5	0.0	0.0	0.0	0	0
84	GRID 6	6	0.0	0.0	0.0	0	0
85	GRID 7	7	0.0	0.0	0.0	0	0
86	GRID 8	8	0.0	0.0	0.0	0	0
87	GRID 9	9	0.0	0.0	0.0	0	0
88	GRID 10	10	0.0	0.0	0.0	0	0
89	GRID 11	11	0.0	0.0	0.0	0	0
90	GRID 12	12	0.0	0.0	0.0	0	0
91	GRID 13	13	0.0	0.0	0.0	0	0
92	GRID 14	14	0.0	0.0	0.0	0	0
93	GRID 15	15	0.0	0.0	0.0	0	0
94	GRID 16	16	0.0	0.0	0.0	0	0
95	GRID 17	17	0.0	0.0	0.0	0	0
96	GRID 18	18	0.0	0.0	0.0	0	0
97	GRID 19	19	0.0	0.0	0.0	0	0
98	GRID 20	20	0.0	0.0	0.0	0	0
99	GRID 21	21	0.0	0.0	0.0	0	0
100	GRID 22	22	0.0	0.0	0.0	0	0
101	GRID 23	23	0.0	0.0	0.0	0	0
102	GRID 24	24	0.0	0.0	0.0	0	0
103	GRID 25	25	0.0	0.0	0.0	0	0

CONVERT

The conversion is not “line by line”. Rather the deck is parsed by identifying each entry type and its continuations line (intelligent storage). Then is internally stored and final conversion is then created (to free, small, or large).

This example combines shows the conversion and also the horizontal lines for each entry (in this case 2 lines per GRID entry).

16csvFull

◀EditConvertf06csvAdvanced

release\input.dat

Band - Actual	Number	Header	☐ None ☐ Small ☒ Large	V Lines	Band	Number
Comments	Style	H Lines	H Lines - Entry	Entry Colors	Entry ID	
	Convert	Run☐ Free ☐ Small ☒ Large ☐ Stored	CommentsWarningsOverwrite Orig			
		Output File: Browse... C:\AView\target\release\output.dat				
		Command: aview-cli.exe "C:\AView\target\release\input.dat" "C:\AView\target\release\output.dat" --type=large				
Advanced		?	Convert Options: ParamRight AlignHeader EmbedHeader Remove			

8--->--9--->--10-->

PSPC

8--->--9--->--10-->

\$--1--->--2--->--3--->--4--->--5--->--6--->

GRID

GRID ID

CID1

X1 (X)

X2 (Y)

X3 (Z)

CID2

PSPC

83\$--1--->--2--->--3--->--4--->--5--->--6--->--7--->--8--->--9--->--10-->

84GRID*

1

0

27.13655

-7.05793*

85*

1.

0

37.13655

-7.05793*

86GRID*

2

0

37.13655

-7.05793*

87*

1.

0

37.13655

-7.05793*

88GRID*

3

0

37.13655

-7.05793*

89*

0.0

0

f06csv

F06csv is a program that intelligentially parses a f06 and creates an internal library. From there, the user can extract data in a very specific manner.

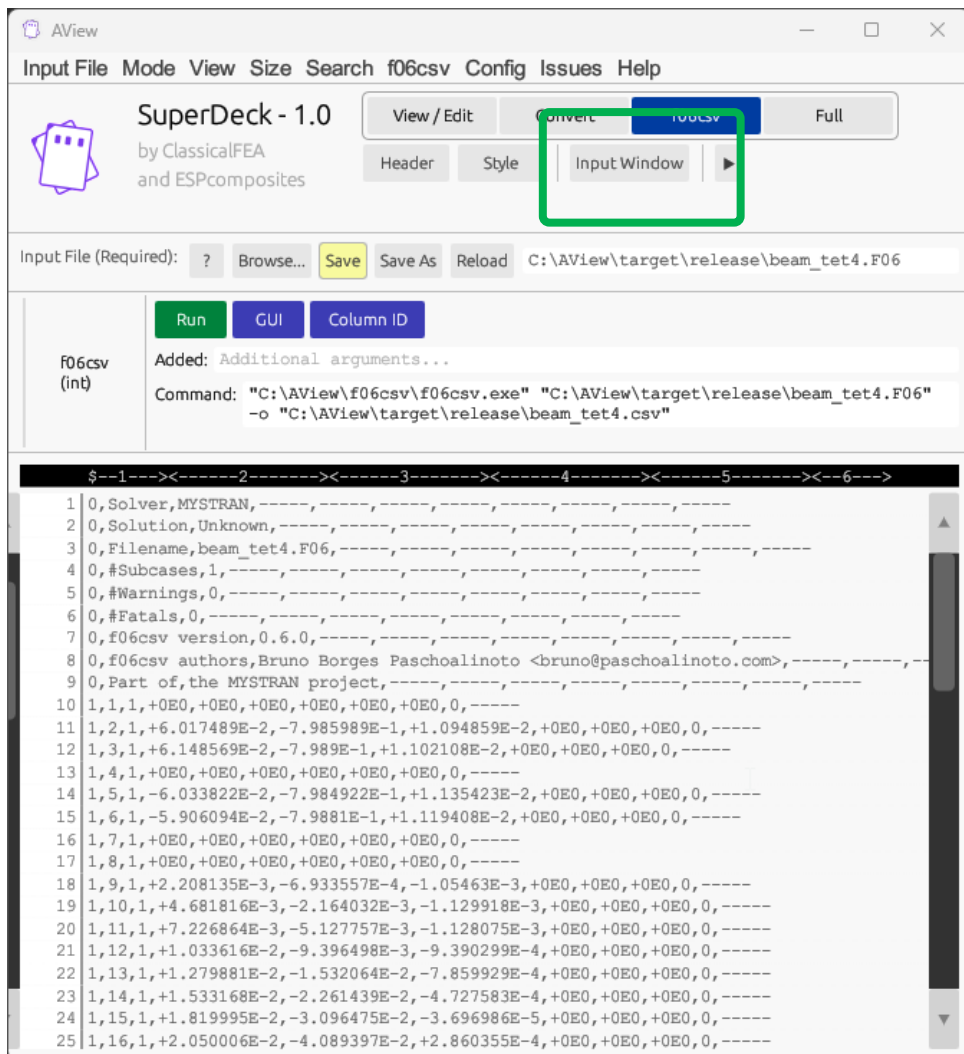
This is simple example with no flags or filters.

The screenshot displays the AView application window. The top menu bar includes 'Input File', 'Mode', 'View', 'Size', 'Search', 'f06csv', 'Config', 'Issues', and 'Help'. Below the menu, the 'SuperDeck - 1.0' logo is visible, along with buttons for 'View / Edit', 'Convert', 'f06csv', and 'Full'. The 'Input File (Required):' field shows the path 'C:\AView\target\release\beam_tet4.F06'. The 'f06csv (n)' section shows the command: 'C:\AView\f06csv\f06csv.exe" "C:\AView\target\release\beam_tet4.F06" -o "C:\AView\target\release\beam_tet4.csv"'. The main window is divided into two panes. The left pane shows a table of data with columns for line number, index, and various numerical values. The right pane shows the output of the f06csv command, which includes a list of parameters and their values, such as 'Solver,MYSTRAN', 'Solution,Unknown', 'Filename,beam_tet4.F06', and 'f06csv version,0.6.0'.

Line	Index	Value
114	4	0.0
115	5	0.0
116	6	0.0
117	7	0.0
118	8	0.0
119	9	0.0
120	10	0.0
121	11	0.0
122	12	0.0
123	13	0.0
124	14	0.0
125	15	0.0
126	16	0.0
127	17	0.0
128	18	0.0
129	19	0.0
130	20	0.0
131	21	0.0
132	22	0.0
133	23	0.0
134	24	0.0
135	25	0.0
136	26	0.0
137	27	0.0
138	28	0.0

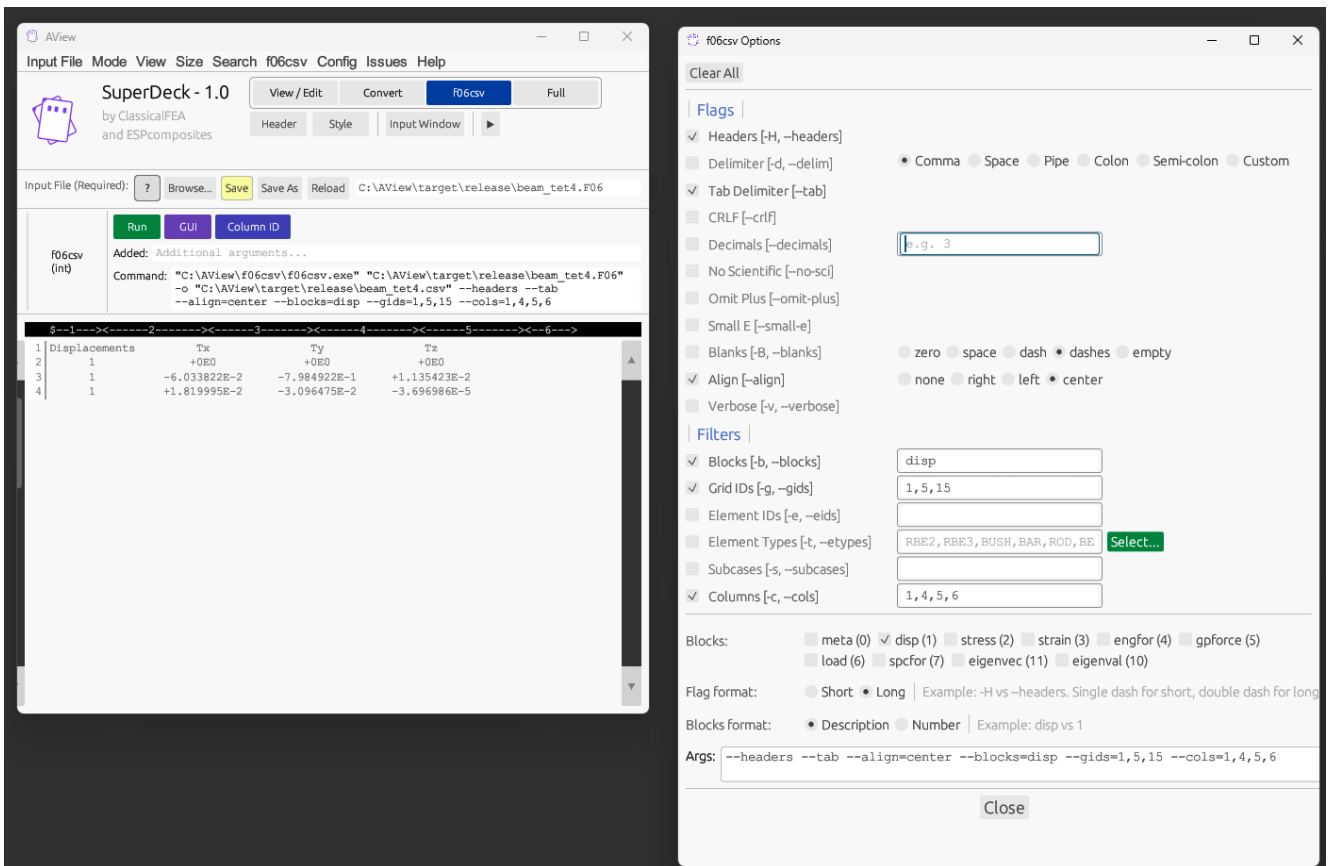
f06csv

It is worth nothing that for both the Convert and f06csv options, the input window (left window) can be hidden. This can be useful for the f06csv GUI interface (next slide).



f06csv

Using the GUI window, all of the f06csv flags and filters are available. This example shows the displacements for specified grids (nodes) for specified DOF. The result is also formatted for improved readability. The default CSV is typically used to read into other programs.



NOTE: The **Run** button is available even when the flyout GUI is shown. This allows the use to quickly see the result of the GUI flyout modifications.

Advanced

There are various advanced features, which can be accessed via the flyout submenu

